MiRQuery Tutorial for new users

1. Uploading your data files

1. **Required**: Raw miRNA counts as a .csv file, in a miRNA (rows) by sample (columns) format, where the column headers are SampleIDs. Users should avoid hyphens, spaces, and other special characters except the underscore (" _ ") or period (" . "). The first column should be miRNA features and should not have a header. Features should be mature miRNA IDs from miRBase v. 22.

Download the example:
[MiRQuery/data/COREAD_Downsampled_Tumor_vs_Normal_Mature_miRNA.csv at main · MSDLLCpapers/MiRQuery]

	TCGA.SS.A7HO.01A	TCGA.D5.6930.01A	TCGA.A6.2684.11A	TCGA.CK.5913.01A	TCGA.AA.3548.01A	TCGA.A6.2671.11A	TCGA.AA.3520.11A	TCGA.D5.5541.01A	TCGA.D5.6538.01A	TCGA.A6.A565.01A	TCGA.A6.2683.11A	TCGA.G4.6586.01A
hsa-let-7a-5p	66146	97802	45053	118912	9809	22257	40456	136741	67181	96053	47618	71656
hsa-let-7b-5p	32054	52861	108274	33368	11374	76243	65727	41669	24959	33064	114411	26651
hsa-let-7c-5p	2370	3203	1684	1059	523	2766	3520	2823	650	579	2416	550
hsa-let-7d-5p	5335	1755	5182	1750	884	5277	4376	1960	2240	2868	6964	2141
hsa-let-7e-5p	6895	4297	749	2410	236	633	901	1705	3081	3260	853	2854
hsa-let-7f-5p	24900	25335	483	67644	1720	205	368	81102	34099	36353	442	45521
hsa-let-7g-5p	4792	1485	245	1934	528	73	215	2706	1986	4236	118	3727
hsa-let-7i-5p	822	2425	485	1083	263	233	438	1853	567	2659	629	1806
hsa-miR-1-3p	9	131	0	18	12	0	0	79	68	125	2	14
hsa-miR-1-5p	6	125	0	11	8	2	1	49	62	149	0	13
hsa-miR-100-5p	1893	23517	1819	1827	817	904	1239	15402	11032	6276	1712	3322
hsa-miR-101-3p	19924	21376	78	4334	2558	90	87	17681	10039	24987	123	7351
hsa-miR-101-5p	19338	21505	91	4183	2465	94	67	17347	9687	24341	118	7088
hsa-miR-103a-2-5p	26223	25025	5859	47899	4323	2086	4404	42040	31790	74389	4960	21110
hsa-miR-103a-3p	26205	24881	5794	47615	4302	2146	4463	41817	31516	74803	5050	21420
hsa-miR-103b	0	0	0	0	0	0	0	0	0	0	0	0
hsa-miR-105-5p	2289	0	9	0	1	6	11	17	3	1	2	0
hsa-miR-106a-5p	201	19	5	19	73	5	3	31	58	69	6	50
hsa-miR-106b-5p	4739	2192	541	1630	371	596	628	2040	2231	3547	1066	2687

2. **Required**: Sample metadata as a .csv file, with each row containing information about each sample. The `SampleID` column is required and must be named as such. Other columns can be any name, but users should avoid hyphens, spaces, and other special characters except the underscore (" _ ") and period (".") in the column header and in the `SampleID`. All SampleIDs in the miRNA counts file should be represented in this file.

Download the example:
[MiRQuery/data/COREAD_Tumor_vs_Normal_Metadata.csv at main · MSDLLCpapers/MiRQuery]

SampleID	bcr_patient_barcode	primary_diagnosis	tissue_type	sex	base_id	Sequencing_Depth	TCGA	barcode
TCGA.AA.3854.01A	TCGA-AA-3854-01A	Mucinous adenocarcinoma	Tumor	female	TCGA-AA-3854	1596769	TCGA-COAD	NA
TCGA.QG.A5YW.01A	TCGA-QG-A5YW-01A	Adenocarcinoma, NOS	Tumor	female	TCGA-QG-A5YW	5250283	TCGA-COAD	NA
TCGA.AA.3966.01A	TCGA-AA-3966-01A	Mucinous adenocarcinoma	Tumor	female	TCGA-AA-3966	2924524	TCGA-COAD	NA
TCGA.G4.6626.01A	TCGA-G4-6626-01A	Adenocarcinoma, NOS	Tumor	male	TCGA-G4-6626	2000675	TCGA-COAD	NA
TCGA.CM.5868.01A	TCGA-CM-5868-01A	Adenocarcinoma, NOS	Tumor	female	TCGA-CM-5868	5882490	TCGA-COAD	NA
TCGA.AA.3973.01A	TCGA-AA-3973-01A	Adenocarcinoma, NOS	Tumor	male	TCGA-AA-3973	2684232	TCGA-COAD	NA
TCGA.AA.3673.01A	TCGA-AA-3673-01A	Adenocarcinoma, NOS	Tumor	female	TCGA-AA-3673	530174	TCGA-COAD	NA

3. **Optional**: Raw gene counts as a .csv file, in a gene (rows) by samples (columns) format. The first column should contain genes in the format of ENSEMBL IDs and should not have a header.

Download the example:
[MiRQuery/data/COREAD_Downsampled_Tumor_vs_Normal_mRNA.csv at main · MSDLLCpapers/MiRQuery]()

	TCGA.SS.A7HO.01A	TCGA.D5.6930.01A	TCGA.A6.2684.11A	TCGA.CK.5913.01A	TCGA.AA.3548.01A	TCGA.A6.2671.11A	TCGA.AA.3520.11A	TCGA.D5.5541.01A	TCGA
ENSG00000000003.15	8315	2165	5334	2981	1936	5865	5977	16687	
ENSG00000000005.6	2	1	44	67	18	28	47	33	
ENSG00000000419.13	3230	1142	1193	1928	569	1348	1833	7123	
ENSG00000000457.14	671	301	746	656	269	747	1118	892	
ENSG00000000460.17	666	263	141	485	262	135	250	600	
ENSG00000000938.13	127	199	220	439	90	194	340	753	
ENSG00000000971.16	286	861	1466	889	230	2239	3278	1815	
ENSG00000001036.14	2440	2828	2665	3267	2191	3252	3846	4007	
ENSG00000001084.13	2532	1235	1568	2146	663	1832	2836	5153	
ENSG00000001167.14	1580	1061	1152	2399	426	1001	1745	3247	
ENSG00000001460.18	522	356	1194	827	241	1568	1480	556	
ENSG00000001461.17	1980	1710	3448	2641	858	4150	4723	2342	
ENSG00000001497.18	3843	2240	1418	2331	1112	1594	2156	5556	
ENSG00000001561.7	1956	881	1565	1801	345	2121	2346	2062	
ENSG00000001617.12	7369	1229	714	762	882	1198	1067	1879	
ENSG00000001626.16	12551	674	9225	10302	1359	9280	15389	9761	
ENSG00000001629.10	2318	1530	1743	2462	467	2231	2954	4445	
ENSG00000001630.17	349	172	198	487	84	213	385	420	
ENSG00000001631.16	327	111	179	177	109	179	259	532	
ENSG00000002016.18	844	221	617	647	119	543	582	579	

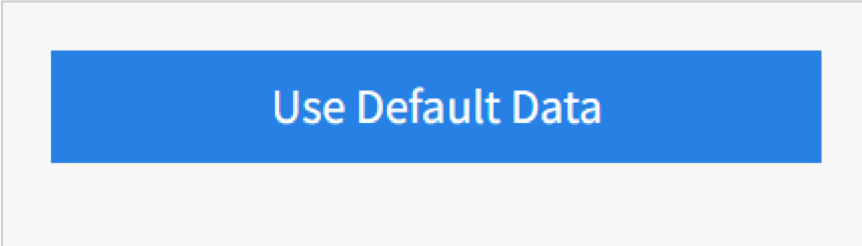
---

2. First-run walkthrough using the demo dataset

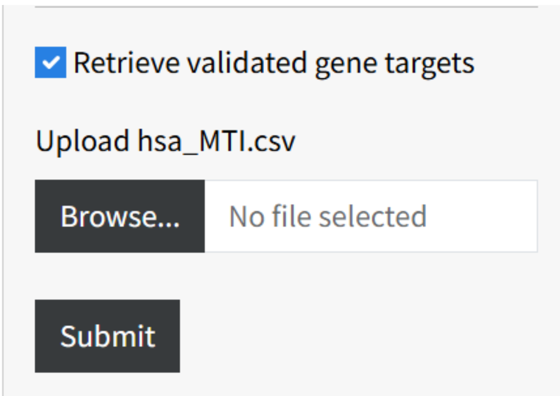
The demo dataset is derived from TCGA colorectal cancer (TCGA-COAD/READ) and includes matched miRNA and mRNA counts plus metadata. It is pre-formatted to meet all MiRQuery requirements.

Step 1: Launch MiRQuery and load the demo data

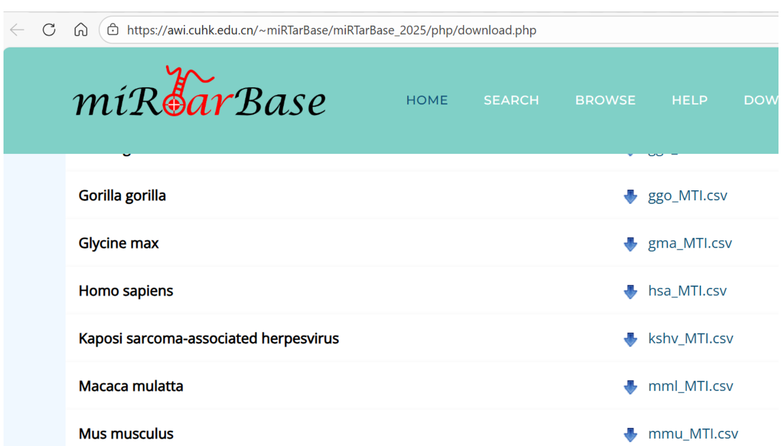
1. Open MiRQuery. We have a hosted instance at [MiRQuery: Interactive Analysis of MicroRNA Sequencing Data]() , or you may follow the instructions on our Github homepage to install it locally.
2. At the top of the sidebar, click the `Use Default Data` button to load our demo dataset.

3. 

4. Confirm that:
    - miRNA counts, metadata, and mRNA counts are loaded.
    - The **species** is correctly set to *Homo sapiens*, for the purpose of this tutorial.
        - MiRQuery currently only supports *Homo sapiens* OR *Mus musculus*.
5. If experimentally validated targets from miRTarBase are desired, please check the box below to receive the prompt for uploading the correct miRTarBase file.



Here is the link to [miRTarBase: the experimentally validated microRNA-target interactions database](), which is also provided in the Quickstart tab. You will need to download the prompted file (human `hsa_MTI.csv` or mouse `mmu_MTI.csv` file), then click "Browse" to upload from your laptop.

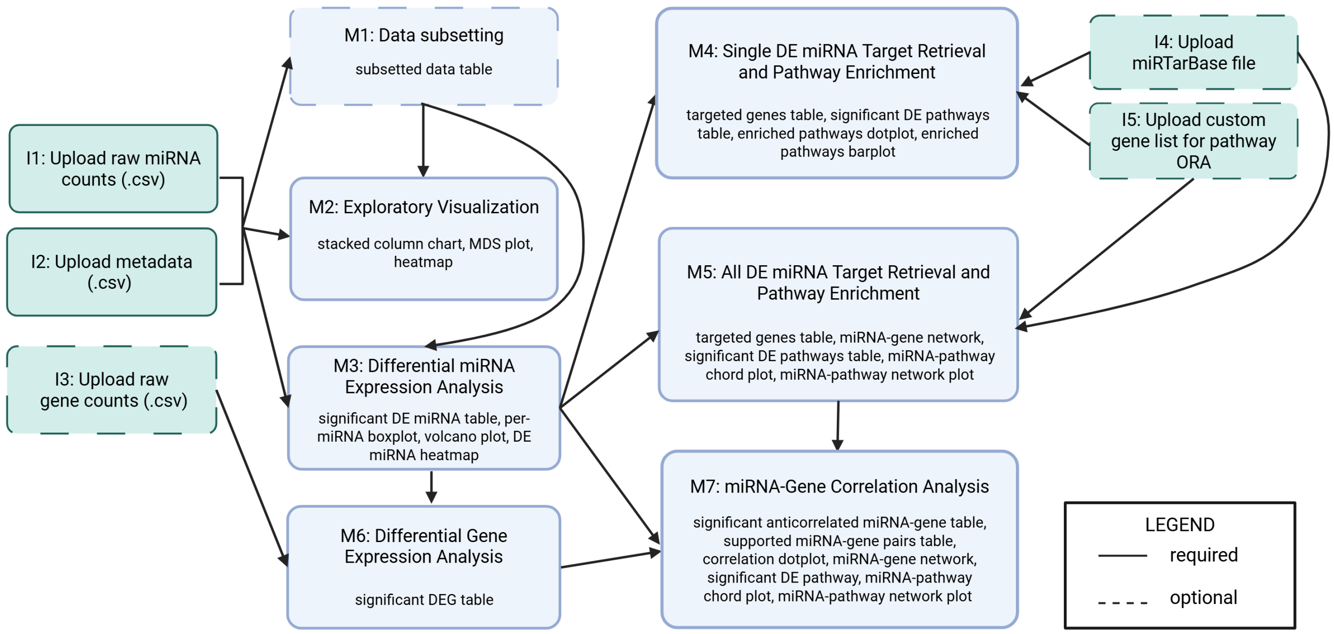


6. Click the `Submit` button.
Once loaded, you can proceed to the analysis modules in the order below.

---

3. Recommended module sequence and expected outputs

MiRQuery is organized into tabs corresponding to each major analysis step. For many analyses, you can move back and forth between modules, but the following sequence is recommended for a first run.



3.1 Exploratory Visualization (M2)

Goal: Understand global patterns in miRNA expression and sample relationships; perform basic quality checks.

1. Go to the **Exploratory Visualization** tab.
2. Select a **grouping variable** from the metadata (for the demo, choose `tissue_type` with levels `Normal` and `Tumor`).

Main outputs and interpretation:

- **Stacked column chart**
  - Shows the relative contribution of each miRNA to total counts per sample.
  - Use the percentage filter to focus on miRNAs that make up, for example, at least 1% of total counts.
  - Look for:
    - Clear differences in dominant miRNAs between groups.
    - Outlier samples with unusual composition patterns.

- **Heatmap of most abundant miRNAs**
  - Uses variance-stabilized counts so that highly abundant miRNAs do not dominate the color scale.
  - Shows expression of up to the top 50 most abundant miRNAs, with optional clustering by samples and miRNAs.
  - Interpretation:
    - Clustering by `tissue_type` suggests that miRNA profiles distinguish normal from tumor.
    - Outlier samples may indicate technical issues or mislabeled samples.
- **Multidimensional scaling (MDS) plot**
  - Visualizes samples in 2D based on miRNA expression distances.
  - Samples that cluster together have similar miRNA profiles.
  - Check whether samples cluster by `tissue_type` or other metadata variables.

Interpretation guidance:

- If samples separate clearly by your selected metadata variable , this supports a strong biological signal.
- If samples do not group as expected, consider checking for mislabeled samples, batch effects, or other confounders before proceeding.

3.2 Differential miRNA Expression Analysis (M3)

Goal: Identify miRNAs that are significantly associated with a variable of interest (e.g. `tissue_type` encoding tumor vs normal).

1. Navigate to the **Differential miRNA Expression Analysis** tab.
2. Specify the **model design**:
   - Select the **primary variable of interest** (e.g. `tissue_type`).
   - Optionally add **covariates** (e.g. `sex`) to adjust for potential confounding and/or reduce unexplained variability that can increase precision of the estimate for the variable of interest (`tissue_type`). Covariates are most important when they are imbalanced across groups or represent known technical factors.
   - Choose the **reference level** (e.g. `Normal` as reference vs `Tumor`). Here, by selecting `Normal` as reference, positive log 2 fold change values will correspond to Tumor-enriched miRNAs, while negative log2 fold change values will correspond to Normal-enriched miRNAs.
3. Run the analysis. MiRQuery uses **DESeq2** to:

- Normalize counts using the median-of-ratios method to account for differences in library size (sequencing depth) and to be robust to a subset of strongly changing miRNAs.
- Fit a negative binomial regression model for each miRNA.
- Estimate log2 fold changes and Benjamini–Hochberg adjusted p-values.

Main outputs and interpretation:

- **DE results table**
  - Includes `log2FoldChange`, raw p-value `pval`, and adjusted p-value (FDR) `padj`.
    - `log2FoldChange` is your effect size, with larger absolute values corresponding to larger effect sizes. Typically, a log2FoldChange of 1 is used as a "biologically meaningful" difference (the gene is 2 times more expressed in tumors compared to normal samples) but it's up to you to decide what is a meaningful log2FoldChange.
  - Default significance filter: adjusted p-value `< 0.05`.
  - Interpretation:
    - Positive `log2FoldChange` indicates higher expression in the non-reference group (e.g. tumor vs normal), which in this case is tumors.
    - Negative `log2FoldChange` indicates lower expression.
- **Boxplots for individual miRNAs**
  - Show normalized counts of a selected differentially expressed miRNA across groups.
    - These counts are normalized by the median of ratios
  - Use these to assess effect size and variability.
- **Volcano plot**
  - x-axis: `log2FoldChange`; y-axis: `-log10(adjusted p-value)`.
  - Highlight miRNAs above a user-specified absolute log2 fold-change threshold.
  - Interpretation:
    - Points far to the right/top: strongly upregulated and significant.
    - Points far to the left/top: strongly downregulated and significant.
- **Heatmap of DEmiRNAs**
  - Displays variance-stabilized counts of significant DEmiRNAs across samples.
    - This normalizes the counts so that highly expressed genes in one sample don't dominate the visual.
  - Clustering can reveal subgroups and patterns of up- vs down-regulation.

Practical guidance:

- For downstream analyses, you may wish to restrict to DEmiRNAs with:
  - Adjusted p-value `< 0.05`, and

- Absolute `log2FoldChange` ≥ 1, depending on your tolerance for weaker changes.

3.3 All DE miRNA Target Retrieval and Pathway Enrichment (M5)

Goal: Move from lists of DEmiRNAs and their target genes to higher-level biological interpretation by testing whether certain pathways or biological processes are overrepresented.

1. Go to **All DE miRNA Target Retrieval and Pathway Enrichment**.
2. Choose the **target type**:
   - **Predicted targets** via miRNAtap (integrating multiple prediction databases):
     - Each database uses rules based on sequence complementarity, conservation, and other properties.
     - If multiple databases agree on a target, that target is often considered more reliable. We recommend an intersection of two.
   - **Validated targets** via miRTarBase:
     - These are based on published experiments (e.g. reporter assays, Western blots) showing that a miRNA regulates a gene.
     - They provide stronger evidence than predictions alone.
3. You can choose to:
   - Use **predicted targets**, **validated targets**, or
   - Combine the two (intersection or union). We recommend choosing the broader union option if you have paired RNA-seq data, so you have more opportunities to identify miRNA-gene pairs with database support, but ultimately this is up to how conservative you want your analysis to be.
4. For the manuscript, we chose predicted gene targets with 2 minimum intersecting sources out of 5.

Statistically, this is not a formal test by itself. It's about defining a plausible set of **candidate genes** regulated by your DEmiRNAs.

Expected behavior:

- This step can be time-consuming, especially with many DEmiRNAs. For our dataset of 372 miRNAs significantly associated with `tissue_type` at a p<0.05, this took 10 minutes. MiRQuery shows progress so you can monitor runtime.
- Once complete, targets are stored for later modules (including correlation analysis).

Outputs and interpretation:

- **miRNA–gene network**

- Nodes: miRNAs and target genes.
- Edges: miRNA-gene associations.
- Filters:
  - Show only nodes with at least *n* connections (minimum degree).
  - Highlight top *n* most connected nodes.
- Interpretation:
  - Highly connected genes may represent central hubs regulated by many DEmiRNAs.
  - Highly connected miRNAs may exert broad regulatory influence.

5. **Pathway overrepresentation analysis (ORA)**
After MiRQuery identifies differentially expressed miRNAs (DEmiRNAs) and retrieves their predicted and/or validated target genes, the next step is to ask whether these target genes **cluster into known biological pathways more than expected by chance**. This is done using **pathway over-representation analysis (ORA)**.

In ORA, each pathway (e.g., a Reactome pathway or a GO term) is treated as a predefined set of genes. MiRQuery tests whether your target-gene list overlaps a pathway **more than would be expected if genes were drawn at random** from a specified **background (universe)**. Statistical significance is assessed using the hypergeometric/Fisher's exact framework as implemented in ***clusterProfiler***, and p-values are adjusted for multiple testing using **Benjamini–Hochberg (BH)** correction (default q-value/FDR cutoff 0.05).

ORA depends on three main choices:

1. **Which miRNAs to analyze:** users can run ORA separately for targets of **upregulated** vs **downregulated** DEmiRNAs to obtain direction-specific biological interpretations.
2. **Which pathway database to use:** MiRQuery supports **Reactome** and **Gene Ontology (GO)**.
3. **Which background universe to use:** the background should represent the set of genes that *could have appeared* in your target list.
   - The background universe choices are:
     - All annotated genes for the organism.
     - This is our default option for predicted target genes. A more target-specific universe (e.g., "all genes that are predicted targets in the underlying prediction resources") would be desirable in principle, because it would restrict the background to genes that are *reachable* by the prediction framework. However, the prediction step in MiRQuery relies on **miRNAtap**, which aggregates predictions from multiple databases via resources stored in precompiled backends (e.g., database files such as SQLite). These backends are not exposed in a way that allows MiRQuery to enumerate the full set of

"eligible" target genes across databases.
- Therefore, MiRQuery uses an annotation-based background ("all annotated genes") as a transparent, stable, and reproducible reference set. Users should interpret ORA results as enrichment relative to the genomic annotation background, and may consider background sensitivity as a limitation when comparing results across different target resources or prediction settings.
- All genes in miRTarBase (if using validated targets).
- This is the best option for when you are retrieving targets of miRNA from miRTarBase, because only the genes in miRTarBase are eligible to be retrieved.
- A user-specified custom list.
- We recommend using this option if your analysis only considers a subset of genes (e.g. if you have RNA-seq data of the same types of samples, use the list of expressed genes, or if you have an external tissue reference expression atlas, and can provide that list of genes). The enrichment question then becomes: Among genes expressed in this samples belonging to this tissue, are predicted targets of my DEmiRNAs overrepresented in certain pathways? A tissue-specific "expressed genes" background can be used when the goal is to interpret predicted miRNA targets in a particular tissue context. This can improve interpretability because genes not expressed in the tissue are unlikely to be functionally targeted. Users should define the expressed-gene list reproducibly (data source, version, and expression cutoff). Results may differ from analyses using all annotated genes because the underlying question changes.

- ORA result row interpretation:
  - **Count**: how many of your genes are in this pathway?
  - **geneID**: which specific genes are they?
  - **GeneRatio**: what fraction of your gene set is in this pathway?
  - **BgRatio**: how common is this pathway among all possible genes (background)?
  - **FoldEnrichment**: how much more enriched is this pathway in your genes vs random expectation?
  - **RichFactor**: what fraction of this pathway's genes are present in your gene set?
  - **zScore**: are the genes in this pathway mainly upregulated (positive) or downregulated (negative), and how strongly?

- **Output and interpretation**
  - Chord plot:
    - Bottom half: miRNAs.
    - Top half: enriched pathways.
    - Ribbon width reflects the extent to which miRNAs target genes in a pathway.
  - Network plot:
    - Nodes: miRNAs (blue) and pathways (purple).

        - Edges: miRNA–pathway associations.
    - Interpretation:
        - Use filters (minimum coverage, number of pathways, node degree) to focus on the most biologically relevant relationships.
        - Look for pathways known to be involved in your disease context (e.g. collagen remodeling, Rho GTPase signaling in colorectal cancer).

3.4 Single DE miRNA Target Retrieval and Pathway Enrichment (M4)

Goal: Explore the functional context of one DEmiRNA of particular interest.

1. Open **Single miRNA Target Retrieval and Pathway Enrichment**.
2. Select a DEmiRNA from the dropdown (e.g. `hsa-miR-21-5p` in the demo dataset).
3. Choose target retrieval options (predicted, validated, or their combination).
4. Run target retrieval and then pathway enrichment (Reactome or GO).

Outputs:

- A table of target genes for the selected miRNA.
- Enriched pathways displayed as **dotplots** or **barplots**, with download options.

Interpretation:

- These plots show which pathways are most strongly associated with that miRNA's targets.
- In the demo, `hsa-miR-21-5p` is associated with Toll-like and interleukin signaling, consistent with known inflammatory roles.

3.5 Differential Gene Expression and miRNA–Gene Correlation Analysis

These steps require the optional mRNA count matrix.

3.5.1 Differential Gene Expression Analysis (M6)

Goal: Identify differentially expressed genes (DEGs) using the same design as for miRNAs.

1. Go to **Differential Gene Expression Analysis**.
2. Run the analysis using DESeq2.

Output:

- Table of DEGs with log2 fold changes and BH-adjusted p-values.
- You can download the full DEG list.

3.5.2 miRNA–Gene Correlation Analysis (M7)

Goal: Identify negatively correlated miRNA-gene pairs that are also supported by target databases.

1. Navigate to **miRNA–Gene Correlation Analysis**.
2. Set a **log2 fold-change threshold** to define which DEmiRNAs and DEGs to use for correlation (e.g. absolute `log2FoldChange` ≥ 1). The DEmiRNAs and DEGs that are passed to this module are those that have an absolute `log2FoldChange` ≥ 0.5.
3. Run correlation analysis:
    - Spearman correlations are computed between all eligible DEmiRNA–DEG pairs.
    - p-values are based on asymptotic t-approximation and corrected using BH.
    - Optional global permutation test:
        - This will run the correlation of all DE miRNA x all DEG meeting your log 2 fold-change threshold X number of times (by default: 100) so it is **very time-intensive!!**
            - Start with a modest number of permutations (e.g., 50–100) and increase if a more stable p-value estimate is needed.
        - In each permutation, MiRQuery shuffles the sample order in the miRNA expression matrix, thereby breaking the pairing between miRNA and mRNA profiles across samples.
        - For each permuted dataset, MiRQuery recalculates pairwise Spearman correlations and BH-adjusted p-values, and records the number of pairs with correlation < 0 and BH-adjusted p-value below the selected cutoff.
        - The observed value is compared with the permutation-based null distribution to calculate a one-sided permutation p-value.
        - **Interpretation.** A small permutation p-value (e.g., < 0.05) indicates a global enrichment of negative miRNA–mRNA associations beyond random expectation. A larger p-value suggests that the overall number of negative correlations could be explained by chance, although individual miRNA-gene pairs may still be biologically relevant.

Outputs and interpretation:

- **Correlation table**
    - Includes correlation coefficient, raw p-value, and adjusted p-value.
    - Focus on **negative correlations** (miRNA up, gene down or vice versa), consistent with canonical miRNA-mediated repression.

- **Database-supported pairs**
  - Subset of miRNA–gene pairs that are both:
    - Negatively correlated, and
    - Present in the target database results from **All DE miRNA Target Retrieval**.
  - These pairs have both statistical and prior knowledge support.
- **Scatterplot**
  - Each point is a miRNA–gene pair.
  - Color/size reflect correlation magnitude and significance.
  - Filters:
    - Minimum correlation magnitude (e.g. |rho| ≥ 0.7 or 0.8).
    - Maximum adjusted p-value.
    - Minimum absolute log2 fold change.
    - Option to restrict to database-supported pairs.
- **miRNA–gene network**
  - Nodes: miRNAs and genes.
  - Edges: significantly anticorrelated, database-supported pairs.
  - Filter by node degree to highlight key miRNAs or genes.

Interpretation caveats:

- Correlation does not prove causality; it supports hypotheses for further validation.
- Strong negative correlation combined with database support is suggestive but not definitive evidence of direct regulation.

3.5.3 Pathway enrichment on anticorrelated pairs

Within the same module, you can perform ORA on the genes in anticorrelated pairs:

1. Choose the **test set**:
   - Genes in database-supported anticorrelated pairs.
     - Genes are included only if they are in miRNA–gene pairs that are both **negatively correlated** in your data, and **supported by a target resource** (predicted and/or validated targets previously retrieved)
     - Use it when you want a more conservative list with stronger biological plausibility; you are prioritizing likely direct miRNA–mRNA regulatory relationships; or, you want fewer false positives at the cost of missing some real but undocumented interactions.
   - All negatively correlated genes.
     - This restricts the gene list to negative miRNA-gene correlations.

        - Use it when you want to capture both **direct and indirect** relationships (e.g., miRNA affects a regulator that affects the gene); your organism/miRNAs may be poorly covered by target databases; or you want a broad overview of processes associated with anticorrelation patterns.
    - Negatively correlated genes meeting an adjusted p-value threshold.
        - This restricts the gene list to negative miRNA-gene correlations that are statistically significant after multiple-testing correction (BH/FDR).
        - Use it when you want an intermediate option: still data-driven, but less noisy than “all negative correlations", or you want to report results with a clear statistical criterion.
2. Choose the **background**:
    - All annotated genes for the organism.
        - Use it when you want a simple, standard baseline, or you want results comparable to many common enrichment workflows.
        - This is likely an overly broad choice for background because it includes many genes that were never realistically eligible for the correlation analysis.
    - All genes eligible for correlation (significant DEGs with absolute log2 fold change ≥ 0.5).
        - Use it when you want enrichment results that are *conditional on* the fact that you already restricted to DEGs and effect-size thresholds, you want the background to match the effective “search space” that generated your test set, or you want to reduce bias from including genes that could not have entered the anticorrelation results. This is the **most conservative** option.
    - Custom gene list.
        - This can make sense if you want a sensitivity analysis (compare custom universe vs MiRQuery universe option, to see what findings are robust to background selection).

Outputs (pathway results table, chord and network plots) mirror those in **3.3 All DE miRNA Target Retrieval and Pathway Enrichment (M5)** and can be interpreted similarly.

---

4. Troubleshooting and FAQ

Q1. The app reports an error when I upload files. What should I check?

- Ensure the **metadata first column** is named exactly `SampleID`.
- Confirm that **sample IDs match** between:
    - miRNA count matrix column headers,
    - mRNA count matrix column headers (if used), and

  - metadata `SampleID` values.
- Check for:
  - Non-integer values in count matrices.
  - Missing values (blank cells).
  - Extra header rows or columns.
- Make sure you selected the correct **species** for your data.

Q2. How long should target retrieval and correlation analysis take?

- Runtime depends on:
  - Number of DEmiRNAs and DEGs.
  - Minimum intersection of prediction databases selected.
- Typical runtimes (approximate, depending on server and dataset size):
  - Target retrieval for tens to a few hundred DEmiRNAs: from seconds to several minutes.
  - Correlation analysis for thousands of genes and hundreds of miRNAs: several minutes or longer.
- The app displays progress for long-running steps. If a job appears stalled for an extended period, verify data size and filters, and consider tightening thresholds.

Q3. I get very few (or no) significant DEmiRNAs/DEGs. What can I do?

- Confirm that:
  - Groups have sufficient sample size.
  - The model is specified correctly (variable of interest and reference level).
- Consider:
  - Checking for strong covariates or batch effects; include relevant covariates in the model when known.
    - MiRQuery does not perform standalone batch correction (e.g., ComBat). However, users can stratify/color samples by batch in exploratory plots (MDS/heatmaps) to assess whether samples cluster by technical variables. For differential expression, batch effects can often be mitigated by including batch as a covariate in the DESeq2 design formula (e.g., ~ batch + tissue_type), which adjusts the statistical model for systematic batch differences.
      Q4. How should I choose the ORA background universe?
- We discuss this in 3.3 and 3.5.3 Pathway enrichment on anticorrelated pairs. Please see those sections.

Q5. Why do some miRNAs or genes not appear in the correlation or target modules?

- They may not meet your **log2 fold-change** or **adjusted p-value** thresholds.

- They may be missing from the prediction databases or miRTarBase.